STEVEN YANG

(647)469-6927 — stevenyang.142857@gmail.com — Website — linkedin

Education

University of Toronto **Sep. 2024 – May 2028 (Expected)**

- *Honours Bachelor of Science*, Data Science Specialist, Computer Science Major
- cGPA: 4.0/4.0
- **Awards:** Dean's List Scholar (2025, 2026) and Woodsworth College Scholarship Fund (2026)
- Relevant Courseworks: Software Design, Software Tools and System Programming, Data Structures and Analysis, Data Science, Probability and Statistics, Linear Algebra, Multivariable Calculus

Research

Student Researcher **Sep. 2026 – Present**

University of Toronto

- Build **AgentLangBench** to evaluate code generation, formal verification, and runtime performance across programming languages for AI agents.
- Surveyed 42 languages and tools for AI agents. Used pinned toolchains and repeatable Linux tests to identify 16 entries that passed all applicable checks without workarounds.
- Audited AI-generated solutions across languages to assess whether their contracts captured the original task requirements.

Experience

Research and Development Intern **May 2026 – August 2026**

NSFOCUS *Beijing, China*

- Built a Python-based **cyber-monitoring** pipeline integrating **Tavily** and **FOFA** APIs to discover, filter, deduplicate, and risk-score suspicious domains and social-media accounts for security investigation.
- Prototyped a client-profile-driven **threat intelligence** workflow for structured event matching, deduplication, and delivery decision using **Deepseek LLM**.
- Analyzed **AI Coding Agent**'s architecture and workflow in a research report.

Projects

NHANES Supplement Use Analysis **2026** *Python, pandas, NumPy, matplotlib, scikit-learn*

- Analyzed NHANES 2021–2023 health and nutrition datasets in a team of 2 to investigate relationships between demographic, dietary, and supplement-use patterns
- Performed large-scale **data cleaning**, **preprocessing**, **feature engineering** across multiple linked survey datasets.
- Modeled supplement use with statistical and **machine learning** methods and visualized predictive factors and model performance.

Question Bank Platform **2026** *Java, OpenAI API, Full-stack*

- Built a full-stack Java application to extract and structure math exam questions from PDFs into a searchable digital database.
- Integrated **OCR** and multimodal **OpenAI Vision models** to parse question text, diagrams, metadata, and page-level images..
- Designed async processing with confidence scoring, manual review flags, PDF segmentation, and structured JSON storage.

GroupFlow **2025** *Java, Swing, MongoDB, JUnit*

- Designed and implemented a collaborative task and group management application using **Clean Architecture**, separating entities, use cases, and UI layers.
- Built core features including group creation, role-based permissions, task assignment, and progress tracking.
- Improved code maintainability and testability through interface-driven design and unit testing with **JUnit**.
- Collaborated in a team environment using **Git-based workflows**, pull requests, and code reviews.

Technical Skills

- **Programming Languages:** Python, Java, C, JavaScript, HTML/CSS, Rust
- **Developer Tools:** Codex, Claude Code, Git/GitHub, Linux, VS Code, IntelliJ, Jupyter, LaTeX, PostgreSQL, Docker, Verus, Dafny
- **Frameworks/Libraries:** NumPy, pandas, matplotlib, scikit-learn, SciPy, statsmodels, TensorFlow